✍©ASCOL CSIT 2070 Batch

Tribhuvan University
Institute of science and Technology
2069
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Bachelor Level / Fourth Year / Eight semester / Science
Computer Science and Information Technology-(CSc-459)
(Geographical Information System)

Full Mark: 60
Pass Mark: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempts any six questions. [10x6=60]

1. Explain the GIS application and its benefits.
2. Why digital mapping is necessary in GIS? Explain with piratical example.
3. Differentiate between geographic phenomena and data modeling.
4. Explain the spatial database design with example.
5. What d you mean by spatial reference? Explain the quality aspects of spatial data?
6. Explain with example of metadata and clearing house.
7. Explain the system analysis and design of GIS.
8. Write short note on **(any two)**
   (a) Map elements
   (b) Spatial relationship
   (c) Geodatabase
   (d) SDI

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Bachelor Level / Fourth Year / Eight semester / Science — Full Mark: 60

Computer Science and Information Technology-(CSc.459) — Pass Mark: 24

(Geographical Information System) — Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempts any six questions. [10x6=60]

1. Define GIS. Describe the role of GIS in a mountainous country like Nepal. Support your answer with a reference of an application. [2+8]
2. What are various coordinate system used in GIS application? Describe them in brief. [2+8]
3. What are raster an vector data? Differentiate between their properties with advantages and disadvantages. [2+8]
4. What are various sources of GIS data? Describe methods of data capture. [2+8]
5. What do you mean by spatial analysis? How do you perform the raster overlay operation using relational operators? Explain with example. [2+8]
6. What is GPS? How does t work? What are various types of errors related to GPS data? [2+4+4]
7. Explain how does a clearing house works? What problems do you face during the spatial data sharing? [5+5]
8. Write short note on **(any two)** [5+5]
   (a) DTM
   (b) Remote Sensing Data
   (c) Open GIS
   (d) Layer

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Tribhuvan University
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2072
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Bachelor Level / Fourth Year / Eight semester / Science
Computer Science and Information Technology-(CSc.459)
(Geographical Information System)

Full Mark: 60
Pass Mark: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempts any six questions. [10x6=60]

1. Explain the history development of GIS. What are the application of GIS?
2. Explain the mapping projection with example.
3. Explain the spatial modeling with example.
4. Explain the GPS and remote sensing with example.
5. Why is spatial analysis important during the GIS implementation? Explain.
6. What do you mean by metadata? Explain the critical factors of spatial data infrastructure (SDI).
7. Explain the web based GIS system with example.
8. Write short note on **(any two)** [5+5]
   (a) Map layers.
   (b) Digital terrain model
   (c) ITRS
   (d) Buffering

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Tribhuvan University
Institute of science and Technology
2073
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Bachelor Level / Fourth Year / Eight semester / Science
Computer Science and Information Technology-(CSc.459)
(Geographical Information System)

Full Mark: 60
Pass Mark: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempts any six questions. [10x6=60]

1. Define GIS as an Information System. How a country like Nepal can benefit from GIS? Describe its application in Agriculture. [2+4+4]
2. What is distortion? Describe various types of Map Projection along with free hand drawing. [2+8]
3. What are raster and vector data? Differentiate between their properties with advantages and disadvantages. [2+8]
4. What are various sources of GIS data? Describe methods of data capture. [2+8]
5. What is a spatial overlay operator? Explain neighborhood functions that are being implemented in GIS software for analysis. [4+6]
6. Define Geo Database. Explain characteristics of a good database design. [2+10]
7. Explain how does a clearing house works? What problems do you face during the spatial data sharing? [5+5]
8. Write short note on **(any two)** [5+5]
   (a) Functional Component of GIS
   (b) UTM
   (c) Map Elements
   (d) Open GIS

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2074

Bachelor Level / Fourth Year / Eight semester / Science — Full Marks: 60

Computer Science and Information Technology — Pass Marks: 24

(Geographical Information System, **CSC-459**) — Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all questions. **[10x6=60]**

1. Write down major difference between general database management system and GIS.

2. Define datum. What does it mean that Nepal lies in UTM Zone 45 N? Explain.

3. What do you mean by Digital Surface Model(DSM)? Explain with example its applications.

4. List down various sources of GIS data. Explain any three of them in brief.

5. List and describe major map components.

6. What location based services have you noticed in your everyday life? Describe the role of GPS in modern world.

7. Write few advantages and disadvantages of using free and open source GIS software tools with some examples.

8. Write short notes on:

   a) Map resolution
   b) SDI

Prepared by: Rameshwor Dhakal